

R Data Types and Transformations with dplyr

PSYC 259: Principles of Data Science

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**How does R data processing help to
minimize mistakes?**

How can R data processing help us minimize mistakes?

- Code *is* documentation
- Transform data with less human intervention (e.g., `read_tsv()` in the homework)
- But coding errors are also common...what can we do to make sure that our code works?
 - Write code that's reproducible (avoiding absolute file paths)
 - Use code to help clean up variable names to make them human-readable
 - Instead of “hard” coding based on position (such as `ds[1, 2]`), filter and select by name and logical conditions
 - Reduce code duplication (copy-paste-tweak) with loops and functions

Data type and transformation tutorial

- Data types and classes in R
- Logical statements
- Introduction to *dplyr*
- Code style



Follow along from the Github repo

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Part 1: Data Types and Classes

Data types and classes

- Allows language to efficiently store data in memory (*type* in R)
 - If a value is an integer (1, 2, 3, 4) storing it as an integer makes calculations easier compared to storing it as a double (2.34542480424624086)
 - integer, double, logical, and character are R types
- Allows language to follow transformation rules (*class* in R)
 - Addition/subtraction for a date follows different rules compared with integers/double
 - “Less than” makes sense when comparing numbers, but not when comparing strings
 - The class of an object tells the language how to handle it

Common data types/classes in R

- Numeric class
 - integer type: 1, 2, 3
 - double type: 1.12124
- Character type/class: "hello"
- Logical type: T/F (TRUE/FALSE)
- Datetime class (type is double)
- Factor class (type is integer)
- Use `typeof()`, `class()`, `str()`, and `glimpse()` to check the types/classes

Functions to test and convert types

- Take a value and convert it into a different class
 - `as.numeric(x)`, `as.character(x)`, `as.date(x)`
 - returns a typed/classed version of x
- Check if something is a particular class
 - `is.numeric()`, `is.character()`
 - returns a logical (T/F)

Data types

```
x <- 1
```

Data types

```
x <- 1  
typeof(x)
```

```
[1] "double"
```

Data types

```
x <- 1  
typeof(x)  
class(x)
```

```
[1] "double"
```

```
[1] "numeric"
```

Data types

```
x <- 1  
typeof(x)  
class(x)  
is.numeric(x)
```

```
[1] "double"
```

```
[1] "numeric"
```

```
[1] TRUE
```

Data types

```
x <- 1  
typeof(x)  
class(x)  
is.numeric(x)  
is.character(x)
```

```
[1] "double"
```

```
[1] "numeric"
```

```
[1] TRUE
```

```
[1] FALSE
```

Data types

```
x <- 1  
typeof(x)  
class(x)  
is.numeric(x)  
is.character(x)  
as.character(x)
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

Data types

```
x <- 1  
typeof(x)  
class(x)  
is.numeric(x)  
is.character(x)  
as.character(x)  
x + 1
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

Data types

```
x <- 1  
typeof(x)  
class(x)  
is.numeric(x)  
is.character(x)  
as.character(x)  
x + 1
```

```
x <- "Hello"
```

```
[1] "double"
```

```
[1] "numeric"
```

```
[1] TRUE
```

```
[1] FALSE
```

```
[1] "1"
```

```
[1] 2
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
x <- "Hello"
typeof(x)
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
x <- "Hello"
typeof(x)
is.numeric(x)
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
as.numeric(x)
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

[1] NA

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
as.numeric(x)
```

```
#x + ", World" #x + ", World"
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

[1] NA

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
as.numeric(x)
#x + ", World" #x + ", World"
paste0(x, ", World")
```

[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

[1] NA

[1] "Hello, World"

R tries to coerce classes/types to do what you ask

- `+` is a function in R (you can actually call it ``+(1, 2)``)
- `"Hello" + ", World"` fails because the `+` function requires numeric arguments
- `paste0` is happy to take numeric arguments and will coerce them to character
- `paste0(1, 2)` returns `"12"`
- What happens if we evaluate `as.numeric(paste0(1,2))`?
- What happens if we evaluate `TRUE + TRUE`?

Built-in logical comparison functions

- Comparisons between values that result in TRUE/FALSE
 - Greater than/less than (>, >=, <, <=)
 - Equals (==)
 - Not equals (!=)
 - Not (!)
 - And (&)
 - Or (|)
 - %in%

Logical comparisons

```
x <- 1
```

Logical comparisons

```
x <- 1  
x > 0
```

[1] TRUE

Logical comparisons

```
x <- 1  
x > 0  
x > 2
```

[1] TRUE

[1] FALSE

Logical comparisons

```
x <- 1  
x > 0  
x > 2  
x == 1
```

[1] TRUE

[1] FALSE

[1] TRUE

Logical comparisons

```
x <- 1  
x > 0  
x > 2  
x == 1  
x != 2
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

Logical comparisons

```
x <- 1  
x > 0  
x > 2  
x == 1  
x != 2  
!(x == x)
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

Logical comparisons

```
x <- 1  
x > 0  
x > 2  
x == 1  
x != 2  
!(x == x)  
"s" == "S"
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

[1] FALSE

Logical comparisons

```
x <- 1
x > 0
x > 2
x == 1
x != 2
!(x == x)
"s" == "S"
1 > 0 | 0 > 1
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

[1] FALSE

[1] TRUE

Logical comparisons

```
x <- 1
x > 0
x > 2
x == 1
x != 2
!(x == x)
"s" == "S"
1 > 0 | 0 > 1
1 > 0 & 0 > 1
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

[1] FALSE

[1] TRUE

[1] FALSE

Logical comparisons

```
x <- 1
x > 0
x > 2
x == 1
x != 2
!(x == x)
"S" == "S"
1 > 0 | 0 > 1
1 > 0 & 0 > 1
```

```
# Element-wise logical statements
```

```
x <- c(-1, 0, 1)
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

[1] FALSE

[1] TRUE

[1] FALSE

Logical comparisons

```
x <- 1
x > 0
x > 2
x == 1
x != 2
!(x == x)
"s" == "S"
1 > 0 | 0 > 1
1 > 0 & 0 > 1
```

```
# Element-wise logical statements
x <- c(-1, 0, 1)
x < 0
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

[1] FALSE

[1] TRUE

[1] FALSE

[1] TRUE FALSE FALSE

Logical comparisons

```
x <- 1
x > 0
x > 2
x == 1
x != 2
!(x == x)
"s" == "S"
1 > 0 | 0 > 1
1 > 0 & 0 > 1
```

```
# Element-wise logical statements
x <- c(-1, 0, 1)
x < 0
x == 0
```

[1] TRUE

[1] FALSE

[1] TRUE

[1] TRUE

[1] FALSE

[1] FALSE

[1] TRUE

[1] FALSE

[1] TRUE FALSE FALSE

[1] FALSE TRUE FALSE

Logical comparisons

```
x <- 1
x > 0
x > 2
x == 1
x != 2
!(x == x)
"s" == "S"
1 > 0 | 0 > 1
1 > 0 & 0 > 1
```

```
# Element-wise logical statements
```

```
x <- c(-1, 0, 1)
x < 0
x == 0
```

```
# Quickly test if a value is contained in a set
```

```
1 %in% x
```

```
[1] TRUE
```

```
[1] FALSE
```

```
[1] TRUE
```

```
[1] TRUE
```

```
[1] FALSE
```

```
[1] FALSE
```

```
[1] TRUE
```

```
[1] FALSE
```

```
[1] TRUE FALSE FALSE
```

```
[1] FALSE TRUE FALSE
```

```
[1] TRUE
```

How do we use logical statements?

- One common way is through the `ifelse()` command?
- `ifelse(LOGICAL STATEMENT, DO IF TRUE, DO IF FALSE)`

```
x <- c(0, 1, 2, 3, NA)
ifelse(NA %in% x, "x contains a missing value", "x does not contain a missing value")
```

```
[1] "x contains a missing value"
```

```
x <- c(0, 1, 2, 3)
ifelse(NA %in% x, "x contains a missing value", "x does not contain a missing value")
```

```
[1] "x does not contain a missing value"
```

Part 2: Data transformation with the dplyr package

Data transformation with the dplyr package

- A toolbox for common data processing/manipulation operations to apply to tibbles
 - `glimpse()` to see the structure of a tibble
 - `arrange()` to sort data by columns
 - `filter()` and `slice()` to subset data by rows
 - `select()` to subset data by columns
 - `rename()` to rename columns
 - `mutate()` to add or change columns (or their values)
 - `summarize()` and `count()` to calculate summary statistics over rows of data
 - `group_by()` to perform operations within subsets of data
 - and many more...

Each dplyr function uses a similar structure

- `function(data, something_to_do_with_columns)`
- In base R, you need to specify the dataset over and over to subset:
`ds[ds$col1 == 1, 0]`
- In dplyr functions, the data argument lets you work with column names without using `$`. The data argument also lets you access the columns directly without using quoted expressions: `filter(data, col1 == 1)`
- Most dplyr functions return the entire tibble back as an output
- Read help pages (e.g., `?filter`) to understand what arguments each function accepts and what each function returns

Arrange (sorting data)

```
library(tidyverse) #loads dplyr
ds <- starwars #loads built-in star wars database
glimpse(ds)
```

Rows: 87

Columns: 14

```
$ name      <chr> "Luke Skywalker", "C-3PO", "R2-D2", "Darth Vader", "Leia Or...
$ height    <int> 172, 167, 96, 202, 150, 178, 165, 97, 183, 182, 188, 180, 2...
$ mass      <dbl> 77.0, 75.0, 32.0, 136.0, 49.0, 120.0, 75.0, 32.0, 84.0, 77.0...
$ hair_color <chr> "blond", NA, NA, "none", "brown", "brown", "grey", "brown", N...
$ skin_color <chr> "fair", "gold", "white", "blue", "white", "light", "light", "...
$ eye_color  <chr> "blue", "yellow", "red", "yellow", "brown", "blue", "blue", "...
$ birth_year <dbl> 19.0, 112.0, 33.0, 41.9, 19.0, 52.0, 47.0, NA, 24.0, 57.0, ...
$ sex        <chr> "male", "none", "none", "male", "female", "male", "female", ...
$ gender     <chr> "masculine", "masculine", "masculine", "masculine", "femini...
$ homeworld  <chr> "Tatooine", "Tatooine", "Naboo", "Tatooine", "Alderaan", "T...
$ species    <chr> "Human", "Droid", "Droid", "Human", "Human", "Human", "Huma...
$ films      <list> <"A New Hope", "The Empire Strikes Back", "Return of the J...
$ vehicles   <list> <"Snowspeeder", "Imperial Speeder Bike">, <>, <>, <>, "Imp...
$ starships  <list> <"X-wing", "Imperial shuttle">, <>, <>, "TIE Advanced x1",...
```

Arrange (sorting data)

```
library(tidyverse) #loads dplyr
ds <- starwars #loads built-in star wars database
arrange(ds, name)
```

```
# A tibble: 87 × 14
  name      height mass hair_color skin_color eye_color birth_year sex  gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Ackbar      180     83 none      brown mot... orange      41  male masculi...
2 Adi Gal...   184     50 none      dark      blue       NA  fema... femin...
3 Anakin ...   188     84 blond     fair      blue      41.9 male masculi...
4 Arvel C...   NA     NA brown     fair      brown      NA  male masculi...
5 Ayla Se...   178     55 none      blue      hazel      48  fema... femin...
6 BB8         NA     NA none      none      black      NA  none masculi...
7 Bail Pr...   191     NA black     tan       brown      67  male masculi...
8 Barriss...   166     50 black     yellow    blue      40  fema... femin...
9 Ben Qua...   163     65 none      grey, gre... orange     NA  male masculi...
10 Beru Wh...   165     75 brown     light     blue      47  fema... femin...
# i 77 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Arrange (sorting data)

```
library(tidyverse) #loads dplyr
ds <- starwars #loads built-in star wars database
arrange(ds, height)
```

```
# A tibble: 87 × 14
  name      height mass hair_color skin_color eye_color birth_year sex gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Yoda        66    17 white      green      brown        896 male masculi...
2 Ratts T...   79    15 none      grey, blue unknown      NA male masculi...
3 Wicket ...   88    20 brown     brown      brown         8 male masculi...
4 Dud Bolt    94    45 none      blue, grey yellow      NA male masculi...
5 R2-D2       96    32 <NA>      white, bl... red         33 none masculi...
6 R4-P17      96    NA none      silver, r... red, blue    NA none feminin...
7 R5-D4       97    32 <NA>      white, red  red         NA none masculi...
8 Sebulba    112    40 none      grey, red  orange      NA male masculi...
9 Gasgano    122    NA none      white, bl... black      NA male masculi...
10 Watto     137    NA black     blue, grey yellow      NA male masculi...
# i 77 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Arrange (sorting data)

```
library(tidyverse) #loads dplyr
ds <- starwars #loads built-in star wars database
arrange(ds, desc(height), mass)
```

```
# A tibble: 87 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Yarael ...   264    NA none       white       yellow      NA    male  mascu...
2 Tarfful     234   136 brown      brown       blue        NA    male  mascu...
3 Lama Su     229    88 none       grey        black        NA    male  mascu...
4 Chewbac...  228   112 brown      unknown     blue        200    male  mascu...
5 Roos Ta...  224    82 none       grey        orange       NA    male  mascu...
6 Grievous    216   159 none       brown, wh... green, y...  NA    male  mascu...
7 Taun We     213    NA none       grey        black        NA    fema... femin...
8 Tion Me...  206    80 none       grey        black        NA    male  mascu...
9 Rugor N...  206    NA none       green       orange       NA    male  mascu...
10 Darth V... 202   136 none       white       yellow      41.9  male  mascu...
# i 77 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Arrange (sorting data)

```
library(tidyverse) #loads dplyr
ds <- starwars #loads built-in star wars database
arrange(ds, eye_color, hair_color)
```

```
# A tibble: 87 × 14
  name      height mass hair_color skin_color eye_color birth_year sex  gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Nien Nu...   160    68 none      grey      black          NA male masculi...
2 Gasgano     122    NA none      white, bl... black          NA male masculi...
3 Kit Fis...   196    87 none      green     black          NA male masculi...
4 Plo Koon    188    80 none      orange    black          22 male masculi...
5 Lama Su     229    88 none      grey      black          NA male masculi...
6 Taun We     213    NA none      grey      black          NA fema... femin...
7 Shaak Ti    178    57 none      red, blue... black          NA fema... femin...
8 Tion Me...   206    80 none      grey      black          NA male masculi...
9 BB8         NA    NA none      none      black          NA none masculi...
10 Greedo     173    74 <NA>      green     black          44 male masculi...
# i 77 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, height < 100)
```

```
# A tibble: 7 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 R2-D2      96    32 <NA>      white, bl... red          33 none masculin...
2 R5-D4      97    32 <NA>      white, red  red          NA none masculin...
3 Yoda       66    17 white      green       brown       896 male masculin...
4 Wicket S...  88    20 brown      brown       brown        8 male masculin...
5 Ratts Ty...  79    15 none       grey, blue  unknown     NA male masculin...
6 Dud Bolt   94    45 none       blue, grey  yellow      NA male masculin...
7 R4-P17     96    NA none       silver, r... red, blue   NA none feminin...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, name == "Yoda")
```

```
# A tibble: 1 × 14
  name height mass hair_color skin_color eye_color birth_year sex gender
<chr> <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Yoda    66    17 white      green      brown        896 male  masculine
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, is.na(hair_color))
```

```
# A tibble: 5 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>    <dbl> <chr> <chr>
1 C-3PO      167    75 <NA>      gold     yellow     112 none masculi...
2 R2-D2       96    32 <NA>      white, bl... red         33 none masculi...
3 R5-D4       97    32 <NA>      white, red  red         NA  none masculi...
4 Greedo     173    74 <NA>      green     black       44 male masculi...
5 Jabba De... 175  1358 <NA>      green-tan... orange     600 herm... masculi...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, height > 100, height < 150)
```

```
# A tibble: 3 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>      <dbl> <chr> <chr>
1 Watto      137    NA black    blue, grey yellow      NA male mascul...
2 Sebulba    112    40 none     grey, red  orange      NA male mascul...
3 Gasgano    122    NA none     white, blue black       NA male mascul...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, eye_color %in% c("blue", "brown"))
```

```
# A tibble: 40 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
  <chr>    <int> <dbl> <chr>      <chr>    <chr>      <dbl> <chr> <chr>
1 Luke Sk...   172    77 blond      fair     blue        19  male masculi...
2 Leia Or...   150    49 brown      light    brown       19  fema... femin...
3 Owen La...   178   120 brown, gr... light    blue       52  male masculi...
4 Beru Wh...   165    75 brown      light    blue       47  fema... femin...
5 Biggs D...   183    84 black      light    brown       24  male masculi...
6 Anakin ...   188    84 blond      fair     blue       41.9 male masculi...
7 Wilhuff...   180    NA auburn, g... fair     blue       64  male masculi...
8 Chewbac...   228   112 brown      unknown  blue      200  male masculi...
9 Han Solo    180    80 brown      fair     brown       29  male masculi...
10 Jek Ton...  180   110 brown      fair     blue       NA   <NA> <NA>
# i 30 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, !(eye_color %in% c("blue", "brown")))
```

```
# A tibble: 47 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
  <chr>    <int> <dbl> <chr>      <chr>    <chr>    <dbl> <chr> <chr>
1 C-3PO      167    75 <NA>        gold     yellow     112  none  mascu...
2 R2-D2       96    32 <NA>        white, bl... red         33  none  mascu...
3 Darth V...   202   136 none        white     yellow     41.9  male  mascu...
4 R5-D4       97    32 <NA>        white, red red         NA  none  mascu...
5 Obi-Wan...   182    77 auburn, w... fair      blue-gray   57  male  mascu...
6 Greedo     173    74 <NA>        green     black       44  male  mascu...
7 Jabba D...  175  1358 <NA>        green-tan... orange     600  herm... mascu...
8 Wedge A...  170    77 brown       fair      hazel       21  male  mascu...
9 Palpati...  170    75 grey        pale      yellow      82  male  mascu...
10 IG-88     200   140 none        metal     red        15  none  mascu...
# i 37 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (subsetting rows by position)

```
slice(ds, 1:5)
```

```
# A tibble: 5 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>    <dbl> <chr> <chr>
1 Luke Sky...  172    77 blond    fair      blue      19   male masculi...
2 C-3PO      167    75 <NA>     gold      yellow    112  none masculi...
3 R2-D2       96    32 <NA>     white, bl... red       33   none masculi...
4 Darth Va...  202   136 none     white     yellow    41.9 male masculi...
5 Leia Org...  150    49 brown    light     brown     19   fema... feminin...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (subsetting rows by position)

```
slice_head(ds, n = 5)
```

```
# A tibble: 5 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>    <dbl> <chr> <chr>
1 Luke Sky...   172    77 blond    fair     blue      19  male masculi...
2 C-3PO        167    75 <NA>     gold     yellow    112 none masculi...
3 R2-D2         96    32 <NA>     white, bl... red       33  none masculi...
4 Darth Va...   202   136 none     white    yellow    41.9 male masculi...
5 Leia Org...   150    49 brown    light    brown     19  fema... femin...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (subsetting rows by position)

```
slice_tail(ds, n = 4)
```

```
# A tibble: 4 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>    <dbl> <chr> <chr>
1 Rey            NA    NA brown     light    hazel      NA fema... femin...
2 Poe Dame...    NA    NA brown     light    brown      NA male  mascu...
3 BB8            NA    NA none      none     black      NA none  mascu...
4 Captain ...    NA    NA none      none     unknown    NA fema... femin...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (subsetting rows by position)

```
slice_sample(ds, n = 2)
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Jek Tono...   180   110 brown      fair        blue        NA <NA> <NA>
2 Owen Lars    178   120 brown, gr... light       blue        52 male masculi...
```

i 5 more variables: homeworld <chr>, species <chr>, films <list>,
vehicles <list>, starships <list>

Slice (subsetting rows by position)

```
slice_sample(ds, n = 2)
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>    <dbl> <chr> <chr>
1 Obi-Wan ...   182    77 auburn, w... fair    blue-gray    57 male masculi...
2 Finis Va...   170    NA blond     fair    blue        91 male masculi...
```

i 5 more variables: homeworld <chr>, species <chr>, films <list>,
vehicles <list>, starships <list>

Slice (subsetting rows by position)

```
slice_min(ds, height, n = 3)
```

```
# A tibble: 3 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex  gender
<chr>    <int> <dbl> <chr>    <chr>    <chr>    <dbl> <chr> <chr>
1 Yoda      66    17 white     green    brown      896 male masculi...
2 Ratts Ty... 79    15 none     grey, blue unknown    NA male masculi...
3 Wicket S... 88    20 brown     brown    brown       8 male masculi...
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Select (subsetting columns)

```
select(ds, name)
```

```
# A tibble: 87 × 1
  name
<chr>
1 Luke Skywalker
2 C-3PO
3 R2-D2
4 Darth Vader
5 Leia Organa
6 Owen Lars
7 Beru Whitesun Lars
8 R5-D4
9 Biggs Darklighter
10 Obi-Wan Kenobi
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, name, height, mass)
```

```
# A tibble: 87 × 3
```

	name <chr>	height <int>	mass <dbl>
1	Luke Skywalker	172	77
2	C-3PO	167	75
3	R2-D2	96	32
4	Darth Vader	202	136
5	Leia Organa	150	49
6	Owen Lars	178	120
7	Beru Whitesun Lars	165	75
8	R5-D4	97	32
9	Biggs Darklighter	183	84
10	Obi-Wan Kenobi	182	77

```
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, c("name", "height", "mass"))
```

```
# A tibble: 87 × 3
```

	name <chr>	height <int>	mass <dbl>
1	Luke Skywalker	172	77
2	C-3PO	167	75
3	R2-D2	96	32
4	Darth Vader	202	136
5	Leia Organa	150	49
6	Owen Lars	178	120
7	Beru Whitesun Lars	165	75
8	R5-D4	97	32
9	Biggs Darklighter	183	84
10	Obi-Wan Kenobi	182	77

```
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, name:eye_color)
```

```
# A tibble: 87 × 6
```

	name <chr>	height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>
1	Luke Skywalker	172	77	blond	fair	blue
2	C-3PO	167	75	<NA>	gold	yellow
3	R2-D2	96	32	<NA>	white, blue	red
4	Darth Vader	202	136	none	white	yellow
5	Leia Organa	150	49	brown	light	brown
6	Owen Lars	178	120	brown, grey	light	blue
7	Beru Whitesun Lars	165	75	brown	light	blue
8	R5-D4	97	32	<NA>	white, red	red
9	Biggs Darklighter	183	84	black	light	brown
10	Obi-Wan Kenobi	182	77	auburn, white	fair	blue-gray

```
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, -(eye_color:starships))
```

```
# A tibble: 87 × 5
  name          height mass hair_color skin_color
  <chr>         <int> <dbl> <chr>    <chr>
1 Luke Skywalker 172    77 blond    fair
2 C-3PO          167    75 <NA>     gold
3 R2-D2           96    32 <NA>     white, blue
4 Darth Vader    202   136 none     white
5 Leia Organa    150    49 brown    light
6 Owen Lars      178   120 brown, grey light
7 Beru Whitesun Lars 165    75 brown    light
8 R5-D4           97    32 <NA>     white, red
9 Biggs Darklighter 183    84 black    light
10 Obi-Wan Kenobi 182    77 auburn, white fair
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, ends_with("color"))
```

```
# A tibble: 87 × 3
  hair_color skin_color eye_color
  <chr>      <chr>      <chr>
1 blond     fair        blue
2 <NA>      gold        yellow
3 <NA>      white, blue red
4 none      white       yellow
5 brown     light       brown
6 brown, grey light       blue
7 brown     light       blue
8 <NA>      white, red  red
9 black     light       brown
10 auburn, white fair       blue-gray
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, contains("_"))
```

```
# A tibble: 87 × 4
  hair_color    skin_color eye_color birth_year
  <chr>         <chr>      <chr>      <dbl>
1 blond        fair        blue        19
2 <NA>         gold        yellow     112
3 <NA>         white, blue red         33
4 none         white       yellow     41.9
5 brown        light       brown       19
6 brown, grey  light       blue        52
7 brown        light       blue        47
8 <NA>         white, red  red         NA
9 black        light       brown       24
10 auburn, white fair        blue-gray   57
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, where(is.numeric))
```

```
# A tibble: 87 × 3
  height mass birth_year
  <int> <dbl>    <dbl>
1    172    77        19
2    167    75       112
3     96    32        33
4    202   136       41.9
5    150    49        19
6    178   120        52
7    165    75        47
8     97    32        NA
9    183    84        24
10   182    77        57
# i 77 more rows
```

Select (subsetting columns)

```
select(ds, where(is.character))
```

```
# A tibble: 87 × 8
```

	name	hair_color	skin_color	eye_color	sex	gender	homeworld	species
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	Luke Skywalker	blond	fair	blue	male	mascu...	Tatooine	Human
2	C-3PO	<NA>	gold	yellow	none	mascu...	Tatooine	Droid
3	R2-D2	<NA>	white, bl...	red	none	mascu...	Naboo	Droid
4	Darth Vader	none	white	yellow	male	mascu...	Tatooine	Human
5	Leia Organa	brown	light	brown	fema...	femin...	Alderaan	Human
6	Owen Lars	brown, gr...	light	blue	male	mascu...	Tatooine	Human
7	Beru Whitesun...	brown	light	blue	fema...	femin...	Tatooine	Human
8	R5-D4	<NA>	white, red	red	none	mascu...	Tatooine	Droid
9	Biggs Darklig...	black	light	brown	male	mascu...	Tatooine	Human
10	Obi-Wan Kenobi	auburn, w...	fair	blue-gray	male	mascu...	Stewjon	Human

```
# i 77 more rows
```

Select needs good column names

```
bad_names <- tibble(id = 1, depression_1 = 1:5, dep2 = 1:5, DEP3 = 1:5, depressed = T)  
select(bad_names, starts_with("dep"))
```

```
# A tibble: 5 × 4  
  depression_1 dep2 DEP3 depressed  
    <int> <int> <int> <lgl>  
1         1     1     1    TRUE  
2         2     2     2    TRUE  
3         3     3     3    TRUE  
4         4     4     4    TRUE  
5         5     5     5    TRUE
```

Select needs good column names

```
bad_names <- tibble(id = 1, depression_1 = 1:5, dep2 = 1:5, DEP3 = 1:5, depressed = T)  
select(bad_names, starts_with("dep") & !matches("depressed"))
```

```
# A tibble: 5 × 3  
  depression_1 dep2 DEP3  
    <int> <int> <int>  
1         1     1     1  
2         2     2     2  
3         3     3     3  
4         4     4     4  
5         5     5     5
```

Select needs good column names

```
good_names <- tibble(id = 1, dep_1 = 1:5, dep_2 = 1:5, dep_3 = 1:5, is_depressed = T)  
select(good_names, starts_with("dep"))
```

```
# A tibble: 5 × 3  
  dep_1 dep_2 dep_3  
  <int> <int> <int>  
1     1     1     1  
2     2     2     2  
3     3     3     3  
4     4     4     4  
5     5     5     5
```

Select needs good column names

```
good_names <- tibble(id = 1, dep_1 = 1:5, dep_2 = 1:5, dep_3 = 1:5, is_depressed = T)  
select(good_names, matches("_[0-9]"))
```

```
# A tibble: 5 × 3  
  dep_1 dep_2 dep_3  
  <int> <int> <int>  
1     1     1     1  
2     2     2     2  
3     3     3     3  
4     4     4     4  
5     5     5     5
```

Select needs good column names

```
good_names <- tibble(id = 1, dep_1 = 1:5, dep_2 = 1:5, dep_3 = 1:5, is_depressed = T)
select(good_names, num_range(prefix = "dep_", range = 1:3))
```

```
# A tibble: 5 × 3
  dep_1 dep_2 dep_3
  <int> <int> <int>
1     1     1     1
2     2     2     2
3     3     3     3
4     4     4     4
5     5     5     5
```

What's going on here?

```
select(ds, name, height, eye_color)
```

```
# A tibble: 87 × 3
  name          height eye_color
  <chr>         <int> <chr>
1 Luke Skywalker 172 blue
2 C-3P0         167 yellow
3 R2-D2          96 red
4 Darth Vader   202 yellow
5 Leia Organa   150 brown
6 Owen Lars     178 blue
7 Beru Whitesun 165 blue
8 R5-D4          97 red
9 Biggs Darklighter 183 brown
10 Obi-Wan Kenobi 182 blue-gray
# i 77 more rows
```

What's going on here?

```
select(ds, name, height, eye_color)
filter(ds, height < 70)
```

```
# A tibble: 87 × 3
```

	name	height	eye_color
	<chr>	<int>	<chr>
1	Luke Skywalker	172	blue
2	C-3PO	167	yellow
3	R2-D2	96	red
4	Darth Vader	202	yellow
5	Leia Organa	150	brown
6	Owen Lars	178	blue
7	Beru Whitesun Lars	165	blue
8	R5-D4	97	red
9	Biggs Darklighter	183	brown
10	Obi-Wan Kenobi	182	blue-gray

```
# i 77 more rows
```

```
# A tibble: 1 × 14
```

	name	height	mass	hair_color	skin_color	eye_color	birth_
	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>	<
1	Yoda	66	17	white	green	brown	

```
# i 5 more variables: homeworld <chr>, species <chr>, films  
# vehicles <list>, starships <list>
```

What's going on here?

```
select(ds, name, height, eye_color)
filter(ds, height < 70)
ds
```

```
# A tibble: 87 × 3
```

	name <chr>	height <int>	eye_color <chr>
1	Luke Skywalker	172	blue
2	C-3PO	167	yellow
3	R2-D2	96	red
4	Darth Vader	202	yellow
5	Leia Organa	150	brown
6	Owen Lars	178	blue
7	Beru Whitesun Lars	165	blue
8	R5-D4	97	red
9	Biggs Darklighter	183	brown
10	Obi-Wan Kenobi	182	blue-gray

```
# i 77 more rows
```

```
# A tibble: 1 × 14
```

	name <chr>	height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>	birth_... <chr>
1	Yoda	66	17	white	green	brown	

```
# i 5 more variables: homeworld <chr>, species <chr>, films  
# vehicles <list>, starships <list>
```

```
# A tibble: 87 × 14
```

	name <chr>	height <int>	mass <dbl>	hair_color <chr>	skin_color <chr>	eye_color <chr>	bi...
1	Luke Sk...	172	77	blond	fair	blue	
2	C-3PO	167	75	<NA>	gold	yellow	
3	R2-D2	96	32	<NA>	white, bl...	red	
4	Darth V...	202	136	none	white	yellow	
5	Leia Or...	150	49	brown	light	brown	
6	Owen La...	178	120	brown, gr...	light	blue	

Reassign the transformations back to the tibble

```
ds <- select(ds, name, height, eye_color)
```

```
ds
```

```
# A tibble: 87 × 3
```

	name <chr>	height <int>	eye_color <chr>
1	Luke Skywalker	172	blue
2	C-3P0	167	yellow
3	R2-D2	96	red
4	Darth Vader	202	yellow
5	Leia Organa	150	brown
6	Owen Lars	178	blue
7	Beru Whitesun Lars	165	blue
8	R5-D4	97	red
9	Biggs Darklighter	183	brown
10	Obi-Wan Kenobi	182	blue-gray

```
# i 77 more rows
```

Reassign the transformations back to the tibble

```
ds <- select(ds, name, height, eye_color)  
ds <- arrange(ds, height, eye_color)
```

```
ds
```

```
# A tibble: 87 × 3
```

	name <chr>	height <int>	eye_color <chr>
1	Yoda	66	brown
2	Ratts Tyerel	79	unknown
3	Wicket Systri Warrick	88	brown
4	Dud Bolt	94	yellow
5	R2-D2	96	red
6	R4-P17	96	red, blue
7	R5-D4	97	red
8	Sebulba	112	orange
9	Gasgano	122	black
10	Watto	137	yellow

```
# i 77 more rows
```

Reassign the transformations back to the tibble

```
ds <- select(ds, name, height, eye_color)
ds <- arrange(ds, height, eye_color)
ds <- filter(ds, height < 70)
```

```
ds
```

```
# A tibble: 1 × 3
  name    height eye_color
<chr>   <int> <chr>
1 Yoda     66 brown
```

Not a good strategy

```
ds <- starwars
```

Not a good strategy

```
ds <- starwars  
ds_name_height_eye_color <- select(ds, name, height, eye_color)
```

Not a good strategy

```
ds <- starwars  
ds_name_height_eye_color <- select(ds, name, height, eye_color)  
ds_sorted <- arrange(ds_name_height_eye_color, height, eye_color)
```

Not a good strategy

```
ds <- starwars
ds_name_height_eye_color <- select(ds, name, height, eye_color)
ds_sorted <- arrange(ds_name_height_eye_color, height, eye_color)
ds_sorted_filtered <- filter(ds_sorted, height < 70)
```

Not a good strategy

```
ds <- starwars
ds_name_height_eye_color <- select(ds, name, height, eye_color)
ds_sorted <- arrange(ds_name_height_eye_color, height, eye_color)
ds_sorted_filtered <- filter(ds_sorted, height < 70)
ds_sorted_filtered
```

```
# A tibble: 1 × 3
  name    height eye_color
<chr>   <int> <chr>
1 Yoda     66 brown
```

Introducing the pipe operator: %>%

```
ds <- starwars
```

```
ds
```

```
# A tibble: 87 × 14
  name      height mass hair_color skin_color eye_color birth_year sex gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
1 Luke Sk...   172    77 blond      fair        blue         19  male masculi...
2 C-3PO       167    75 <NA>       gold        yellow       112  none masculi...
3 R2-D2        96    32 <NA>       white, bl... red          33  none masculi...
4 Darth V...  202   136 none       white       yellow       41.9  male masculi...
5 Leia Or...  150    49 brown      light       brown        19  fema... feminin...
6 Owen La...  178   120 brown, gr... light       blue         52  male masculi...
7 Beru Wh...  165    75 brown      light       blue         47  fema... feminin...
8 R5-D4        97    32 <NA>       white, red  red          NA   none masculi...
9 Biggs D...  183    84 black      light       brown        24  male masculi...
10 Obi-Wan...  182    77 auburn, w... fair        blue-gray    57  male masculi...
# i 77 more rows
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Introducing the pipe operator: %>%

```
ds <- starwars  
ds <- ds %>% select(name, height, eye_color)
```

```
ds
```

```
# A tibble: 87 × 3  
  name          height eye_color  
  <chr>         <int> <chr>  
1 Luke Skywalker    172 blue  
2 C-3PO             167 yellow  
3 R2-D2              96 red  
4 Darth Vader       202 yellow  
5 Leia Organa       150 brown  
6 Owen Lars         178 blue  
7 Beru Whitesun Lars 165 blue  
8 R5-D4              97 red  
9 Biggs Darklighter  183 brown  
10 Obi-Wan Kenobi    182 blue-gray  
# i 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars  
ds <- ds %>% select(name, height, eye_color)
```

```
ds <- ds
```

```
ds
```

```
# A tibble: 87 × 3
```

	name <chr>	height <int>	eye_color <chr>
1	Luke Skywalker	172	blue
2	C-3PO	167	yellow
3	R2-D2	96	red
4	Darth Vader	202	yellow
5	Leia Organa	150	brown
6	Owen Lars	178	blue
7	Beru Whitesun Lars	165	blue
8	R5-D4	97	red
9	Biggs Darklighter	183	brown
10	Obi-Wan Kenobi	182	blue-gray

```
# i 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds <- ds %>%
  select(name, eye_color)

ds
```

```
# A tibble: 87 × 2
  name                eye_color
  <chr>               <chr>
1 Luke Skywalker     blue
2 C-3PO              yellow
3 R2-D2              red
4 Darth Vader        yellow
5 Leia Organa        brown
6 Owen Lars          blue
7 Beru Whitesun Lars blue
8 R5-D4              red
9 Biggs Darklighter  brown
10 Obi-Wan Kenobi     blue-gray
# i 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars  
ds <- ds %>% select(name, height, eye_color)
```

```
ds <- ds %>%  
  select(name, eye_color) %>%  
  arrange(eye_color)
```

```
ds
```

```
# A tibble: 87 × 2  
  name      eye_color  
  <chr>    <chr>  
1 Greedo   black  
2 Nien Nunb black  
3 Gasgano  black  
4 Kit Fisto black  
5 Plo Koon black  
6 Lama Su  black  
7 Taun We  black  
8 Shaak Ti black  
9 Tion Medon black  
10 BB8     black  
# i 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds <- ds %>%
  select(name, eye_color) %>%
  arrange(eye_color) %>%
  filter(eye_color == "blue")

ds
```

```
# A tibble: 19 × 2
  name                eye_color
  <chr>               <chr>
1 Luke Skywalker     blue
2 Owen Lars          blue
3 Beru Whitesun Lars blue
4 Anakin Skywalker   blue
5 Wilhuff Tarkin     blue
6 Chewbacca          blue
7 Jek Tono Porkins   blue
8 Lobot              blue
9 Mon Mothma         blue
10 Qui-Gon Jinn       blue
11 Finis Valorum      blue
12 Ric Olié           blue
13 Adi Gallia         blue
14 Mas Amedda         blue
15 Cliegg Lars        blue
16 Luminara Unduli    blue
17 Barriss Offee      blue
18 Jocasta Nu         blue
19 Tarfful            blue
```

Assigning the result of a piped expression

All the options below are equivalent, but I stick with `ds <- ds %>%` because it's the most common to see

```
ds <- ds %>% select(height) %>% slice_tail(n = 5)
ds %>% select(height) %>% slice_tail(n = 5) -> ds
ds <- select(ds, height) %>% slice_tail(n = 5)
ds <- slice_tail(select(ds, height), n = 5)
```

Rename

- Using rename can help clean up messy names from other sources
 - Avoid names that contain spaces or start with numbers
 - Force names to use a similar format like snake_case or camelCase
- Installing and loading the **janitor** package opens up some helpful renaming options

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris %>% rename(sepal_length = Sepal.Length)
```

	sepal_length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris %>% rename(sepal_length = Sepal.Length, sepal_width = Sepal.Width, petal_length = Petal.Length)
```

	sepal_length	sepal_width	petal_length	petal_width	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris %>% rename_with(toupper)
```

	SEPAL.LENGTH	SEPAL.WIDTH	PETAL.LENGTH	PETAL.WIDTH	SPECIES
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris %>% rename_with(tolower, starts_with("Petal"))
```

	Sepal.Length	Sepal.Width	petal.length	petal.width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris %>% clean_names()
```

	sepal_length	sepal_width	petal_length	petal_width	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Rename columns

```
library(janitor)
# Switching to built-in iris data set since starwars has good names
iris %>% clean_names("lower_camel")
```

	sepalLength	sepalWidth	petalLength	petalWidth	species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa
11	5.4	3.7	1.5	0.2	setosa
12	4.8	3.4	1.6	0.2	setosa
13	4.8	3.0	1.4	0.1	setosa
14	4.3	3.0	1.1	0.1	setosa
15	5.8	4.0	1.2	0.2	setosa
16	5.7	4.4	1.5	0.4	setosa
17	5.4	3.9	1.3	0.4	setosa
18	5.1	3.5	1.4	0.3	setosa
19	5.7	3.8	1.7	0.3	setosa
20	5.1	3.8	1.5	0.3	setosa
21	5.4	3.4	1.7	0.2	setosa
22	5.1	3.7	1.5	0.4	setosa
23	4.6	3.6	1.0	0.2	setosa
24	5.1	3.3	1.7	0.5	setosa
25	4.8	3.4	1.9	0.2	setosa
26	5.0	3.0	1.6	0.2	setosa
27	5.0	3.4	1.6	0.4	setosa

Mutate and summarize

- Mutate adds/modifies columns; dataset remains the same size
- Summarize collapses the data set down (by default to 1 row) to calculate summary statistics
- Both functions use a similar form:
 - `mutate(data, new_variable = expression)`
 - `summarize(data, mean = mean(var))`
- Like all other dplyr functions, they return tibbles, so for mutate to "stick" it needs to be assigned back to itself:
 - `data <- mutate(data, var = round(var))`
 - `data <- mutate(data, var_r = round(var))`
- Since summarize collapses to a new level, you might not want to assign it to overwrite the original data set!

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
```

```
ds
```

```
# A tibble: 87 × 4
```

	name <chr>	mass <dbl>	height <int>	hair_color <chr>
1	Luke Skywalker	77	172	blond
2	C-3PO	75	167	<NA>
3	R2-D2	32	96	<NA>
4	Darth Vader	136	202	none
5	Leia Organa	49	150	brown
6	Owen Lars	120	178	brown, grey
7	Beru Whitesun Lars	75	165	brown
8	R5-D4	32	97	<NA>
9	Biggs Darklighter	84	183	black
10	Obi-Wan Kenobi	77	182	auburn, white

```
# i 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = "yes")
```

```
ds
```

```
# A tibble: 87 × 5
```

	name <chr>	mass <dbl>	height <int>	hair_color <chr>	in_movie <chr>
1	Luke Skywalker	77	172	blond	yes
2	C-3PO	75	167	<NA>	yes
3	R2-D2	32	96	<NA>	yes
4	Darth Vader	136	202	none	yes
5	Leia Organa	49	150	brown	yes
6	Owen Lars	120	178	brown, grey	yes
7	Beru Whitesun Lars	75	165	brown	yes
8	R5-D4	32	97	<NA>	yes
9	Biggs Darklighter	84	183	black	yes
10	Obi-Wan Kenobi	77	182	auburn, white	yes

```
# i 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = "yes")
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi))
```

ds

A tibble: 87 × 7

	name <chr>	mass <dbl>	height <int>	hair_color <chr>	in_movie <chr>	height_m <dbl>	bmi <dbl>
1	Luke Skywalker	77	172	blond	yes	1.72	26
2	C-3PO	75	167	<NA>	yes	1.67	27
3	R2-D2	32	96	<NA>	yes	0.96	35
4	Darth Vader	136	202	none	yes	2.02	33
5	Leia Organa	49	150	brown	yes	1.5	22
6	Owen Lars	120	178	brown, grey	yes	1.78	38
7	Beru Whitesun Lars	75	165	brown	yes	1.65	28
8	R5-D4	32	97	<NA>	yes	0.97	34
9	Biggs Darklighter	84	183	black	yes	1.83	25
10	Obi-Wan Kenobi	77	182	auburn, white	yes	1.82	23

i 77 more rows

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = "yes")
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi)) %>%
  arrange(desc(bmi))
```

ds

A tibble: 87 × 7

	name <chr>	mass <dbl>	height <int>	hair_color <chr>	in_movie <chr>	height_m <dbl>	bmi <dbl>
1	Jabba Desilijic Tiure	1358	175	<NA>	yes	1.75	443
2	Dud Bolt	45	94	none	yes	0.94	51
3	Yoda	17	66	white	yes	0.66	39
4	Owen Lars	120	178	brown, grey	yes	1.78	38
5	R2-D2	32	96	<NA>	yes	0.96	35
6	IG-88	140	200	none	yes	2	35
7	R5-D4	32	97	<NA>	yes	0.97	34
8	Jek Tono Porkins	110	180	brown	yes	1.8	34
9	Grievous	159	216	none	yes	2.16	34
10	Darth Vader	136	202	none	yes	2.02	33

i 77 more rows

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = "yes")
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi)) %>%
  arrange(desc(bmi))
```

```
ds <- ds
```

```
ds
```

```
# A tibble: 87 × 7
```

	name	mass	height	hair_color	in_movie	height_m	bmi
	<chr>	<dbl>	<int>	<chr>	<chr>	<dbl>	<dbl>
1	Jabba Desilijic Tiure	1358	175	<NA>	yes	1.75	443
2	Dud Bolt	45	94	none	yes	0.94	51
3	Yoda	17	66	white	yes	0.66	39
4	Owen Lars	120	178	brown, grey	yes	1.78	38
5	R2-D2	32	96	<NA>	yes	0.96	35
6	IG-88	140	200	none	yes	2	35
7	R5-D4	32	97	<NA>	yes	0.97	34
8	Jek Tono Porkins	110	180	brown	yes	1.8	34
9	Grievous	159	216	none	yes	2.16	34
10	Darth Vader	136	202	none	yes	2.02	33

```
# i 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = "yes")
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi)) %>%
  arrange(desc(bmi))

ds <- ds %>%
  filter(hair_color %in% c("blond", NA))

ds
```

A tibble: 8 × 7

	name <chr>	mass <dbl>	height <int>	hair_color <chr>	in_movie <chr>	height_m <dbl>	bmi <dbl>
1	Jabba Desilijic Tiure	1358	175	<NA>	yes	1.75	443
2	R2-D2	32	96	<NA>	yes	0.96	35
3	R5-D4	32	97	<NA>	yes	0.97	34
4	C-3PO	75	167	<NA>	yes	1.67	27
5	Luke Skywalker	77	172	blond	yes	1.72	26
6	Greedo	74	173	<NA>	yes	1.73	25
7	Anakin Skywalker	84	188	blond	yes	1.88	24
8	Finis Valorum	NA	170	blond	yes	1.7	NA

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = "yes")
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi)) %>%
  arrange(desc(bmi))

ds <- ds %>%
  filter(hair_color %in% c("blond", NA)) %>%
  mutate(hair_color = ifelse(is.na(hair_color), "no hair", hair_color),
         mass = ifelse(mass > 1000, "huge", "not huge"))
```

ds

A tibble: 8 × 7

	name <chr>	mass <chr>	height <int>	hair_color <chr>	in_movie <chr>	height_m <dbl>	bmi <dbl>
1	Jabba Desilijic Tiure	huge	175	no hair	yes	1.75	443
2	R2-D2	not huge	96	no hair	yes	0.96	35
3	R5-D4	not huge	97	no hair	yes	0.97	34
4	C-3PO	not huge	167	no hair	yes	1.67	27
5	Luke Skywalker	not huge	172	blond	yes	1.72	26
6	Greedo	not huge	173	no hair	yes	1.73	25
7	Anakin Skywalker	not huge	188	blond	yes	1.88	24
8	Finis Valorum	<NA>	170	blond	yes	1.7	NA

Tricky, common task: Changing some but not all values within a column

```
# Change all NA values to "no hair", keep all others the same  
ds <- ds %>% mutate(hair_color = ifelse(is.na(hair_color), "no hair", hair_color))  
  
# Set all heights greater than 100 to 100, keep all others the same  
ds <- ds %>% mutate(height = ifelse(height > 100, 100, height))
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species)
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species)
ds %>% summarize(min_height = min(height))
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         NA
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species)
ds %>% summarize(min_height = min(height))
ds %>% summarize(min_height = min(height, na.rm = T))
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         NA
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         66
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species)
ds %>% summarize(min_height = min(height))
ds %>% summarize(min_height = min(height, na.rm = T))
```

```
ds %>% summarize(min_height = min(height, na.rm = T),
                 m_height = mean(height, na.rm = T),
                 max_height = max(height, na.rm = T))
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         NA
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         66
```

```
# A tibble: 1 × 3
  min_height m_height max_height
  <int>      <dbl>      <int>
1         66     175.        264
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species)
ds %>% summarize(min_height = min(height))
ds %>% summarize(min_height = min(height, na.rm = T))

ds %>% summarize(min_height = min(height, na.rm = T),
                 m_height = mean(height, na.rm = T),
                 max_height = max(height, na.rm = T))

ds %>% group_by(species)
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         NA
```

```
# A tibble: 1 × 1
  min_height
  <int>
1         66
```

```
# A tibble: 1 × 3
  min_height m_height max_height
  <int>      <dbl>      <int>
1         66      175.        264
```

```
# A tibble: 87 × 4
# Groups:   species [38]
   name          mass height species
  <chr>      <dbl>  <int>  <chr>
1 Luke Skywalker    77    172 Human
2 C-3PO             75    167 Droid
3 R2-D2             32     96 Droid
4 Darth Vader     136    202 Human
5 Leia Organa      49    150 Human
6 Owen Lars       120    178 Human
7 Beru Whitesun Lars  75    165 Human
8 R5-D4            32     97 Droid
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species)
ds %>% summarize(min_height = min(height))
ds %>% summarize(min_height = min(height, na.rm = T))

ds %>% summarize(min_height = min(height, na.rm = T),
                 m_height = mean(height, na.rm = T),
                 max_height = max(height, na.rm = T))

ds %>% group_by(species) %>%
  summarize(min_height = min(height, na.rm = T),
            m_height = mean(height, na.rm = T),
            max_height = max(height, na.rm = T),
            n = n())
```

```
# A tibble: 1 × 1
  min_height
  <int>
```

```
1      NA
```

```
# A tibble: 1 × 1
  min_height
  <int>
```

```
1      66
```

```
# A tibble: 1 × 3
  min_height m_height max_height
  <int>      <dbl>      <int>
```

```
1      66      175.      264
```

```
# A tibble: 38 × 5
  species    min_height m_height max_height     n
  <chr>      <int>      <dbl>      <int> <int>
1 Aleena         79         79         79     1
2 Besalisk      198        198        198     1
3 Cerean        198        198        198     1
4 Chagrian      196        196        196     1
5 Clawdite      168        168        168     1
6 Droid         96        131.        200     6
7 Dug          112        112        112     1
8 Ewok          88         88         88     1
9 Geonosian     183        183        183     1
```